

RESTAPI AI Agent

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Build an **LLM-powered AI Agent** that can dynamically consume a **Swagger/OpenAPI** specification and **invoke REST APIs**, the feature set can be divided into multiple categories.

1. API Discovery & Understanding

Swagger/OpenAPI Ingestion

- Import Swagger/OpenAPI URL
- Upload Swagger JSON/YAML file
- Auto-refresh API definitions
- Support OpenAPI 2.0 and 3.x
- Multiple Swagger sources
- Version management

API Analysis

- Parse endpoints automatically
- Understand request/response schemas
- Detect authentication requirements
- Extract business descriptions
- Generate API summaries
- Build API dependency graph

API Catalog

- Search APIs using natural language
- Categorize APIs
- Tag-based filtering
- Group by business domain
- API documentation viewer

2. Natural Language to API Execution

Intent Understanding

User says:

"Create a customer named John"

Agent automatically identifies:

POST /customers

Features

- Intent detection
- Endpoint selection
- Parameter extraction
- Missing parameter identification
- Context-aware API selection
- Multi-turn clarification

Example:

User:

Create a customer

Agent:

What is the customer's email address?

3. Dynamic API Invocation

Request Generation

- Generate REST requests dynamically
- Populate query parameters
- Populate path parameters
- Populate headers
- Populate request body

Supported Methods

- GET
- POST
- PUT
- PATCH
- DELETE
- OPTIONS

Response Processing

- Parse JSON
- Parse XML
- Parse CSV
- Parse text responses
- Structured output generation

4. Authentication & Security

Authentication Types

- API Key
- OAuth2
- JWT
- Basic Auth
- Bearer Token
- Client Credentials
- Azure AD
- SAML Integration

Security Features

- Secret vault integration
- Credential encryption
- Token refresh
- Role-based access
- Endpoint authorization

5. Agentic Planning & Reasoning

Instead of invoking one API, agent can execute workflows.

Example:

User:

Create a customer and place an order.

Agent Plan:

Step 1: Search Customer

Step 2: Create Customer

Step 3: Get Product

Step 4: Create Order

Step 5: Send Confirmation

Features

- Task decomposition
- Multi-step planning
- Chain API calls
- Dependency resolution
- Reflection and retry
- Goal-oriented execution

6. Workflow Orchestration

Workflow Builder

- Visual workflow designer
- Drag-drop API actions
- Conditions
- Loops
- Parallel execution
- Human approval nodes

Workflow Features

- Reusable workflows
- Scheduled execution
- Event-driven execution
- Workflow templates

7. Context & Memory

Session Memory

- Conversation history
- Previous API responses
- User preferences
- Entity tracking

Long-Term Memory

- Customer information
- Frequently used APIs
- Business context
- Past actions

8. Human-in-the-Loop

Approval Framework

Before executing:

DELETE Customer 123

Agent asks:

This action is irreversible. Approve?

Features

- Approval workflows
- Multi-level approvals
- Escalation matrix
- Audit approvals

9. Error Handling & Recovery

Automatic Recovery

- Retry mechanisms
- Timeout handling
- Circuit breakers
- Alternate API selection

Intelligent Error Resolution

Example:

```
{  
  "error": "Customer Not Found"  
}
```

Agent can:

Search Customer

Create Customer

Retry Order

10. API Testing Capabilities

Testing Features

- Swagger validation
- Endpoint testing
- Schema validation
- Mock responses
- Contract testing

Test Generation

- Generate test cases
- Generate sample payloads
- Generate Postman collections
- Generate curl commands

11. Data Transformation

Transformation Features

- JSON to JSON mapping
- XML to JSON
- CSV processing
- Schema conversion

Example:

CRM Response
to

ERP Request
automatically.

12. AI Capabilities

LLM Features

- Prompt engineering
- Tool calling
- Function calling
- Reflection
- Self-correction
- Multi-agent collaboration

RAG Integration

- Swagger documentation RAG
- Knowledge base RAG
- API usage examples
- Business SOP retrieval